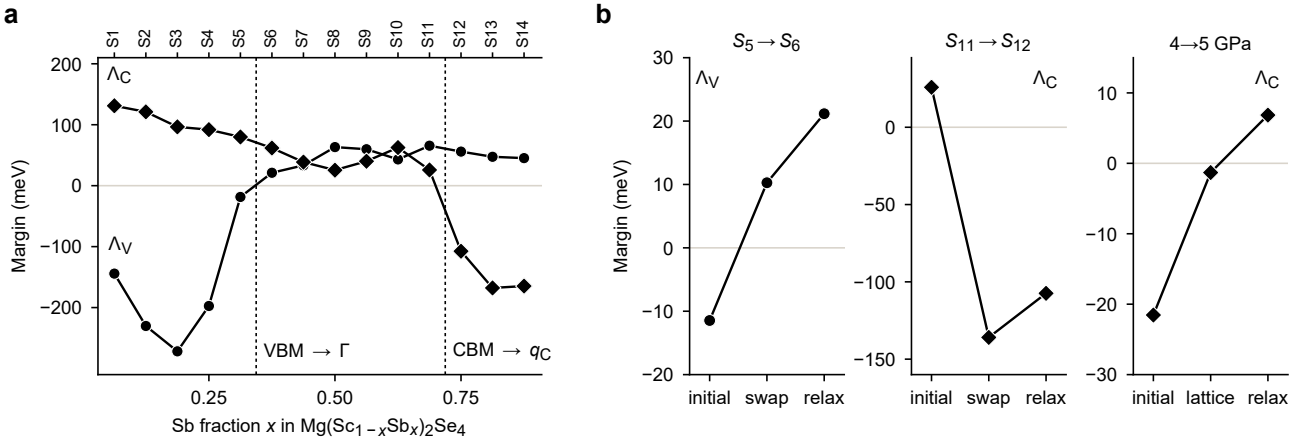

a
S1
S2
S3
S4
S5
S6
S7
S8
S9
S10
S11
S12
S13
S14
200
100
0
−100
−200
Margin (meV)
Λc
Λv
VBM → Γ
CBM → qc
0.25
0.50
0.75
Sb fraction x in Mg(Sc1 − xSbx)2Se4
b
S5 → S6
S11 → S12
4→5 GPa
Λv
Λc
Λc
30
20
10
0
−10
−20
−50
−100
−150
−30
initial
swap
relax
initial
swap
relax
initial
lattice
relax